The law of refraction is also known as Snell's law.

The refraction at a boundary is shown in Figure 6.15.

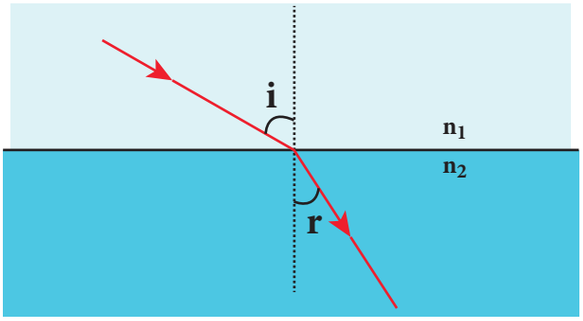
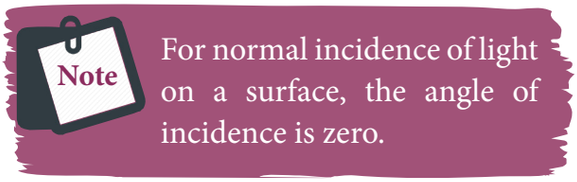


Figure 6.15 Refraction of light



For normal incidence of light on a surface, the angle of incidence is zero.

6.4.1 Angle of deviation due to refraction

The angle between the direction of incident ray and the refracted ray is called *angle of deviation due to refraction.* When light travels from rarer to denser medium, it deviates towards normal as shown in Figure 6.16. The angle of deviation in this case is,

$$d = i - r \qquad (6.20)$$

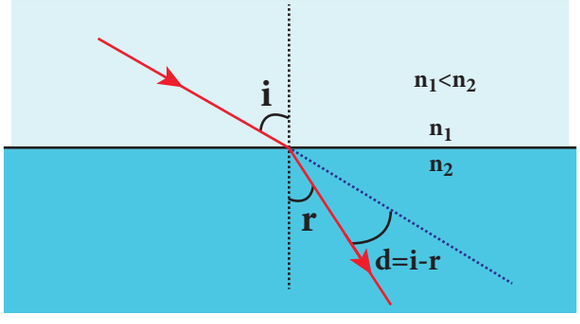


Figure 6.16 Angle of deviation due to refraction from rarer to denser medium

On the other hand, if light travels from denser to rarer medium, it deviates away from normal as shown in Figure 6.17. The angle of deviation in this case is,

$$d = r - i \qquad (6.21)$$

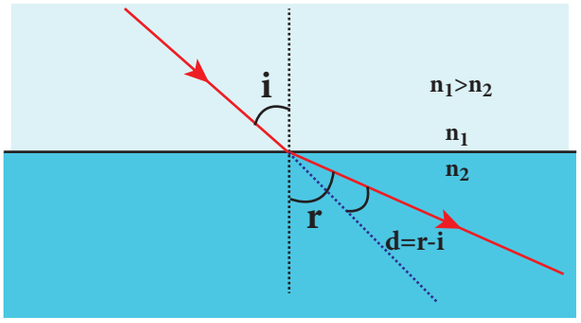


Figure 6.17 Angle of deviation due to refraction from denser to rarer medium

6.4.2 simultaneous reflection (or) refraction

In any refracting surface there will also be some reflection taking place. Thus, the intensity of refracted light will be lesser than the incident light. **The phenomenon in which a part of light from a source undergoing reflection and the other part of light from the same source undergoing refraction at the same surface is called *simultaneous reflection* (or) *simultaneous refraction.*** This is shown in Figure 6.18. Such surfaces are available as partially silvered glasses.

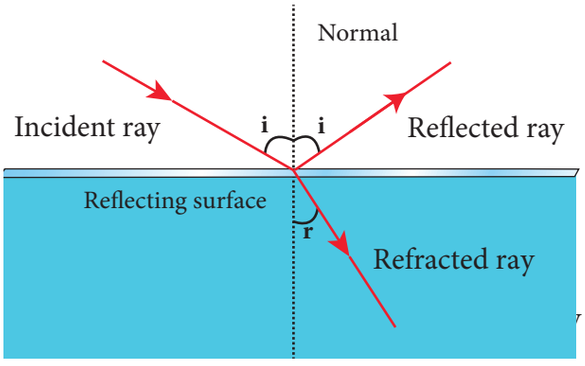


Figure 6.18 Simultaneous reflection (or) refraction

XII_Physics_Vol-2_UNIT-6_EM.indd 15 26-12-2022 11:37:39